

Predictive Modeling for Budgetary Cost Estimation of Wastewater Treatment Plant Projects

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I. PROJECT OVERVIEW

A. BACKGROUND

The preparation of technical and budgetary proposals for wastewater treatment plant projects requires coordination between the design and sales department. The design team is responsible for developing the technical proposals, including process design, equipment selection, bill of quantities, and technical specifications; which are then used by the commercial team to prepare the project cost and final quotation. Keep note that this process requires sufficient time to complete both the technical and commercial aspects of a proposal.

In the usual setup, clients may provide strict deadlines for proposal submissions, particularly for projects involving bidding activities or scheduled presentations. In some cases, the design team may have less than a week to complete the technical proposal, leaving the remaining time for the commercial team to prepare the project cost. The limited time available for both departments can place significant pressure on the proposal preparation process.

On the other hand, budgetary inquiries may initially only require an estimated project cost to determine whether a proposed wastewater treatment project is financially feasible. With this, a complete technical proposal is not necessary since the client is only inquiring about the estimated cost before deciding whether to proceed with further technical discussions or detailed design.

B. PROBLEM STATEMENT-

The preparation process can result in considerable waiting time for clients because a budgetary cost is generally developed after the technical design and equipment requirements have been confirmed. As a result, clients who are only requesting budgetary estimates may need to wait for the completion of the technical design and commercial pricing process. This creates an obstacle when multiple projects have to be evaluated within short deadlines, especially during the third and fourth quarter of the year. Therefore, the requirement to prepare technical proposals before obtaining the exact budgetary estimate can contribute to workload for both the design and commercial teams.

C. PROJECT OBJECTIVE

This project aims to develop a data-driven approach for predicting wastewater treatment plant project costs to support budgetary estimation and improve the efficiency of the technical and commercial proposal process with the following objectives:

1. **To collect, clean, preprocess, and analyze historical wastewater treatment plant project data** in order to identify relevant project parameters associated with overall project cost.
2. **To develop and evaluate Machine Learning Models** for predicting wastewater treatment plant project costs
3. **To validate the predicted project costs** by comparing model predictions with actual historical project costs using appropriate statistical and predictive performance metrics.
4. **To evaluate the potential of the proposed predictive model as a decision-support tool** for reducing the time required for budgetary cost estimation.

D. PROJECT FRAMEWORK

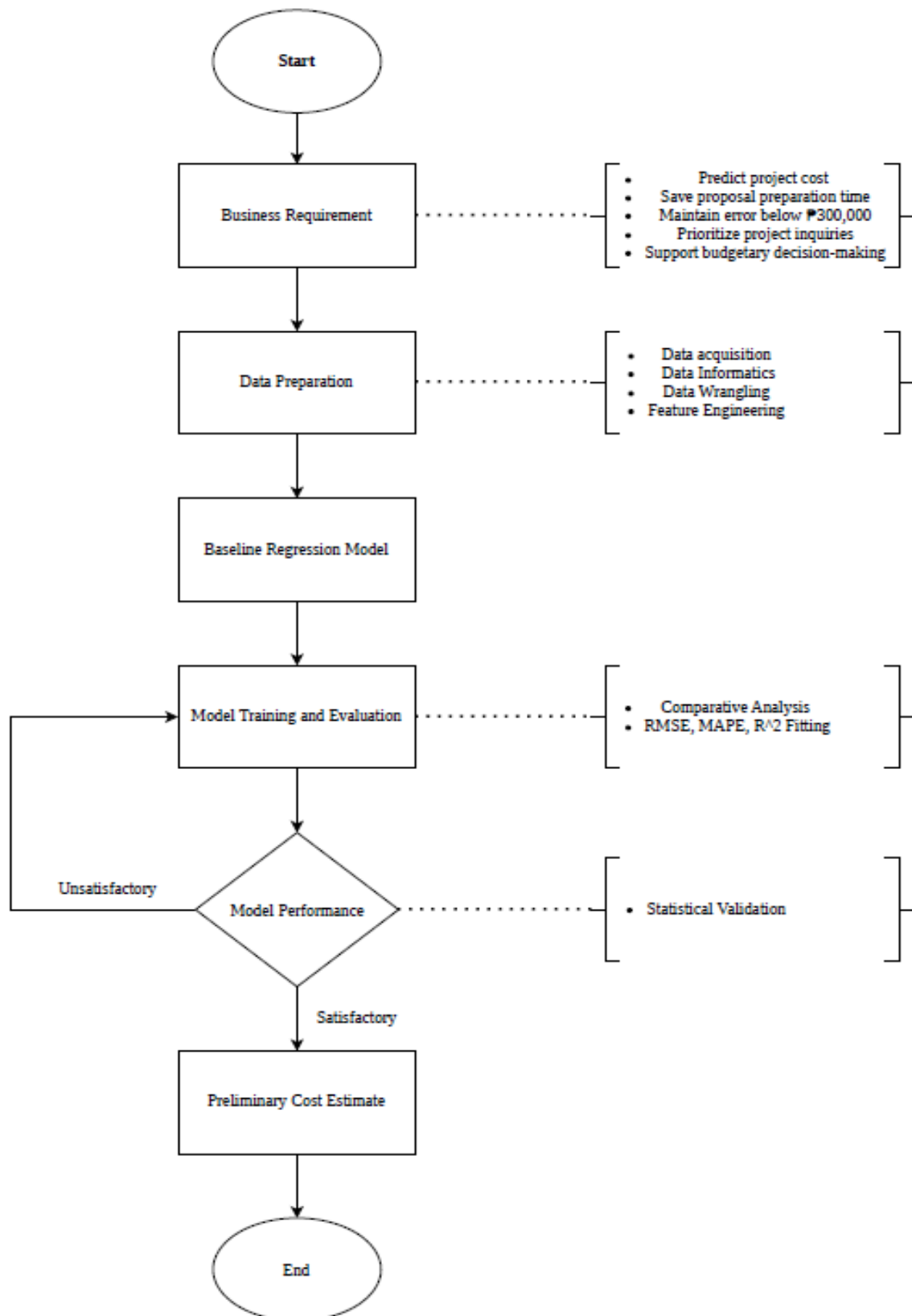


Figure 1.1 Project Framework

II. DATA INFORMATICS

DATA SOURCE

The dataset consists of various wastewater treatment plant project proposals between 2025 and 2026. The data includes key project parameters such as flow rate (expressed in cubic meters per day or CMD), Biological Oxygen Demand (BOD), number of Sequencing Batch Reactors (SBR), and proposed project cost.

DATA FEATURES

1. **Flow Rate** - The client's proposed plant capacity in cubic meters per day.
2. **BOD** - The primary design metric for tank sizing and design configuration.
3. **No. of SBR** - Dictates the number of cycles per day, the number of SBR usually depends on the capacity of the plant and the proposed area.
4. **Region** - Proposed place of the plant
5. Price of the Design

III. DATA WRANGLING

EXPLORATORY DATA ANALYSIS

A. CORRELATION

According to figure 3.1, the correlation matrix shows a strong positive relationship between flow rate and project cost, with a correlation coefficient of 0.95. This relationship is actually aligned with the expectations, as projects with higher wastewater flow rates generally require greater treatment capacity.



Figure 3.1 Correlation Matrix of Features

Furthermore, the Number of SBR displays a moderate positive correlation with both CMD (0.46) and Price (0.51) because the reactor count is highly dependent on land availability and operational cycles; while a single SBR tank might suffice for four cycles a day under ideal conditions, a restricted site area forces engineers to deploy multiple smaller SBR units to keep individual tank sizes structurally manageable without over-dimensioning them.

B. DISTRIBUTION ANALYSIS

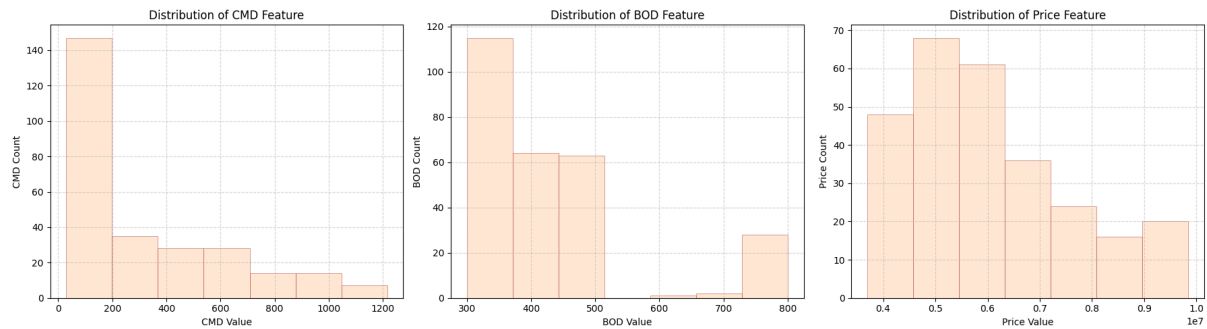


Figure 3.2 Distribution Analysis of Features

The CMD feature shows that most proposed projects fall within the 50-200 CMD range, indicating that a large portion of the projects in the dataset involve relatively small wastewater flows. This distribution reflects the frequency of smaller-scale developments and industries with small wastewater effluent discharge. CMD values exceeding 200 CMD represent projects with greater wastewater treatment requirements, such as larger hotels with higher water consumption from commercial kitchens and other operations, while the highest CMD values may correspond to large-scale residential developments.

The BOD feature reflects the variation in wastewater characteristics among the different project types. Most BOD values are concentrated within the 300 - 500 mg/L range, which generally correspond to domestic wastewater characteristics. Moreover, higher BOD concentrations, particularly higher than 600 mg/L range, is associated with commercial developments with kitchen and laundry wastes which can contribute to higher organic loading.

The Price output variable is consistent with the observed distribution and correlation to CMD. Project costs are predominantly and concentrated toward lower values, reflecting the frequency of small-scale projects in the dataset.

C. OUTLIER ANALYSIS

With the help of distribution analysis, the right-skewed distribution indicates that high-cost projects are not statistical outliers; rather, the current dataset is primarily concentrated on small-scale developments.

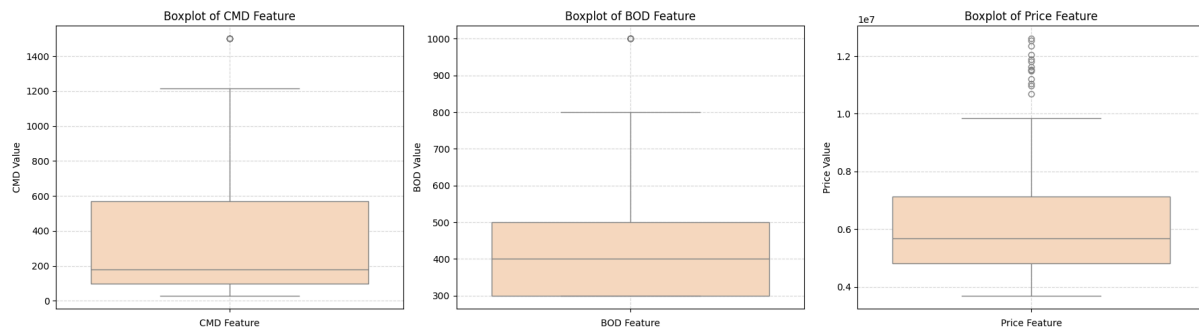


Figure 3.3 Outlier Analysis of Features

OUTLIER TREATMENT

Given the data constraints, two approaches to handling outliers will be investigated:

1. Outlier removal: Identify and remove high-cost outliers from the dataset before model development.
2. Outlier retention and treatment: Retain the high-cost observations and investigate data treatment techniques, including K-nearest neighbors (KNN), where applicable.

D. FEATURE ENGINEERING

For the Region feature, One-Hot Encoding was applied to convert the categorical region data into numerical variables suitable for machine learning. The data set consists of seven regions: NCR, CALABARZON, MIMAROPA, Central Luzon, North Luzon, Visayas, and Mindanao. Region was included as a predictor because project pricing may vary depending on location. The pricing department applies location contingency costs to account for transportation, logistics, mobilization, and other regional factors.

IV. MODEL EVALUATION

A. ERROR METRICS

This project utilized the use of 3 statistical metrics, Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and coefficient of determination (R^2) as expressed in equations 1 - 3, respectively.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_p - y_m)^2} \quad (1)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{A_i - F_i}{A_i} \right| \times 100 \quad (2)$$

$$R^2 = 1 - \frac{\sum_{I=1}^n (y_p - y_m)^2}{\sum_{I=1}^n (y_p - \overline{y_m})^2} \quad (3)$$

Where n is the population or the total number of inputs, y_m is the actual values, y_p is the predicted values, and $\overline{y_m}$ is the mean of the observed values for a given input parameter.

B. COMPARATIVE ANALYSIS

Comparative analysis was conducted using machine learning methods. The machine learning models evaluated in this project included Linear Regression (LR), Decision Tree Regressor (DTR), Random Forest Regressor (RFR), XGBoost Regressor (XGBR). The performance of these models was evaluated using error metrics to identify which model demonstrated the best predictive capability.

Furthermore, two handling of outlier methods were also evaluated: KNN, and dropping the higher-scale projects. In table 4.1 and 4.2, the capability of the models was assessed based on the error metrics. Amongst the three models, **dropping the outliers yielded the least error**; therefore, the results obtained using this method will be used for the discussion.

Table 4.1 Summary of Results: Outlier Treatment using KNN.

Machine Learning Predictive Models			
Models	RMSE	MAPE	R ²
LR	₹ 470,412.01	6.58 %	0.9084
DTR	₹ 495,904.12	6.64 %	0.8918
RFR	₹ 386,175.50	5.21 %	0.9344
XGBR	₹ 313,431.51	3.51 %	0.9568

Table 4.2 Summary of Results: Dropping Higher-Scale Project Outliers.

Machine Learning Predictive Models			
Models	RMSE	MAPE	R ²
LR	₹ 428,002.20	5.84 %	0.9084
DTR	₹ 380,775.97	5.42 %	0.9121
RFR	₹ 372,542.54	5.61 %	0.9159
XGBR	₹ 70,031.88	0.87 %	0.9970

Based on the results in Table 4.2, the XGBR model produced the lowest RMSE of ₹ 70,031.88 and the highest R² score of 0.9970 amongst the evaluated machine learning models, this indicates the strongest predictive performance after the removal of high-scale project outliers. The R² scores of the different machine learning models used for cost prediction were ranked from highest to lowest as follows: XGBR, RFR, DTR, and LR. Moreover, the RMSE scores from highest to lowest: XGBR, RFR, DTR, and LR. Therefore, **XGBR demonstrated the best model for predicting cost.**

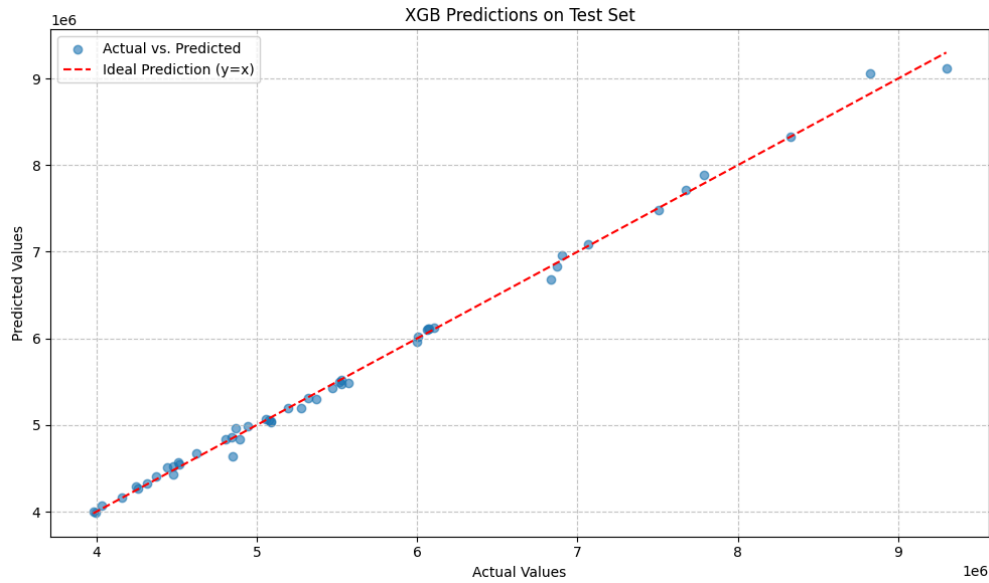


Figure 4.1 R^2 Fit of XGB Model

The test set and prediction scatter plot shows that the XGBoost model achieves good fit on the test set. As illustrated in figure 4.1, the low variance of points indicates highly correlated between the true and predicted values, demonstrating reliable predictions. Furthermore, even though the dataset is somehow sparse in high-scale projects the model still exhibits stability in predicting high prices.

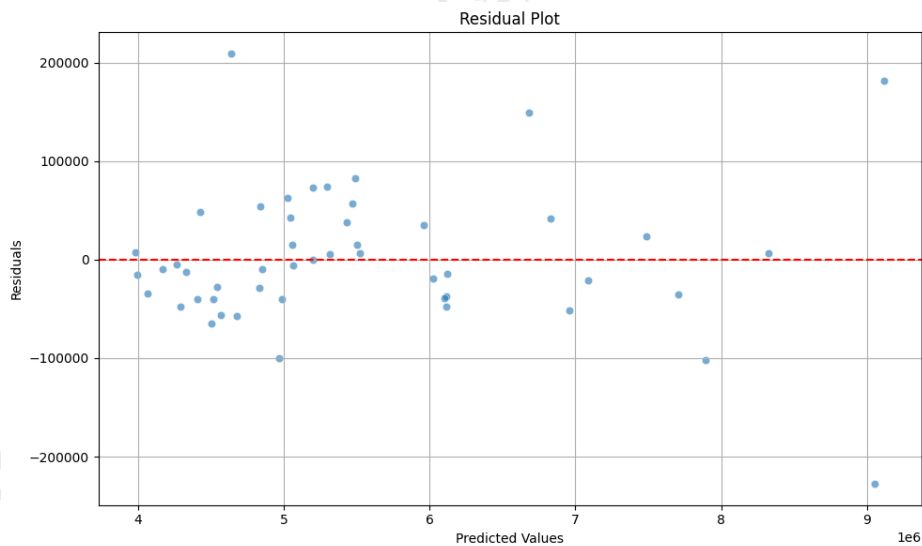


Figure 4.2 Residual Plot of XGB Model

Residual plot was used to graphically analyze the distribution of errors and check for hidden patterns in the model's prediction. Based on figure 4.2, the errors are randomly distributed above and below the perfect prediction line. However, few outliers are visible near the boundaries of the data, especially around 4.5 million to 9 million. Because of this further testing should be considered which will be discussed in the next chapter.

V. RESULTS AND DISCUSSION

A. FEATURE IMPORTANCE

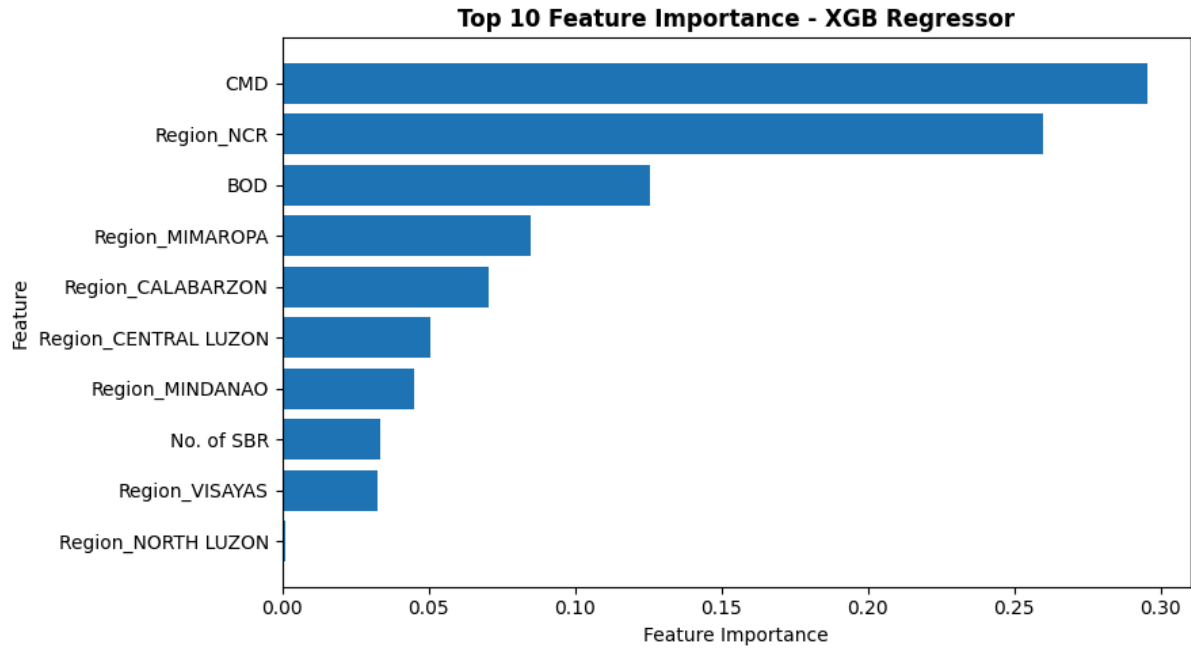


Figure 5.1 Feature Importance of XGBRegressor Model

Comparing the linear correlation matrix with a tree-based feature importance highlights how the model utilizes linear and non-linear relationships. Both correlation matrix and feature importance agree on the significance of CMD in predicting the design cost, from a linear correlation score of 0.95 to the highest feature score of 0.29. When it comes to the number of SBR tanks, from a linear perspective it displays a score of 0.51 while in the XGB Regressor model it was one of the lowest predicting functions amongst other features.

The most valuable insight from this comparison comes from BOD. In the correlation matrix, BOD shows a near-zero linear relationship with Price (0.026). However, the XGBoost model ranks BOD as its third most important feature (0.12). This direct contradiction suggests that BOD has a highly non-linear or interactive relationship with the target variable.

B. BLAND - ALTMAN ANALYSIS

Implementing a Bland-Altman analysis provides a validation step for the residual plot analysis because it formalizes the exact error behaviours observed by establishing limits of agreement. While the residual plot shows that errors are generally centered around zero, a Bland-Altman plot maps the error differences against the mean of the actual and predicted values, directly revealing if the model's accuracy degrades at specific scales.

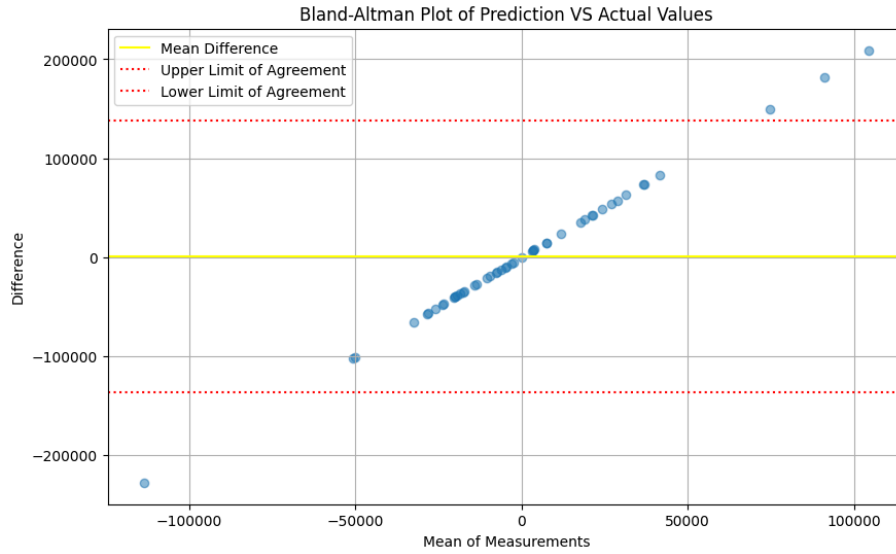


Figure 5.2 Bland-Altman Plot of XGB Regressor Model

The observed differences between the prediction and the actual value was close to zero, indicating minimal error in the model predictions. Furthermore, the observed differences lie between the upper and lower limit of agreement; therefore, the residuals of the XGB Regressor model lie within the acceptable range.

The plot clearly shows several significant outliers at the extreme ends of the plot, with an extreme negative deviation plunging below -200,000 and multiple positive deviations breaking well past the +140,000 upper limit of agreement. Even though extreme outliers were actively dropped during the final preprocessing phase residual outliers still occurred.

C. HYPOTHESIS TESTING

Table 5.1. Hypotheses and Significant Level

Null Hypothesis (H_0)	The model cannot accurately predict the design cost, and there is no statistically significant relationship between the input features and the final cost
Alternative Hypothesis (H_1)	The model can accurately predict the design cost, and there is a statistically significant relationship between the input features and the final cost.
Significance Level (α)	0.05

A paired t-test was used to analyze the model predictions and the actual project cost. This test determines whether the mean difference between the two sets of values is statistically different from zero. The results of the test provide an objective basis for assessing whether the XGBoost model produces predictions that are statistically comparable to the actual design costs. Moreover, the hypothesis test was conducted using a 5% significance level. Since the p-value of 0.9972 is greater than the significance level of 0.05, the null hypothesis is not rejected. Therefore, **there is no statistically significant difference between the predicted design costs and the actual cost results**, indicating that the predictive model is statistically aligned with the true measurements. On the other hand, the t-statistic of 0.003485, being exceptionally close to zero, indicates that **the mean difference between the actual and predicted design cost values is minimal**.

VI. CONCLUSION

In conclusion, this project showed the potential of machine learning as a data-driven approach for preliminary budgetary cost estimation of wastewater treatment plant projects. Among the evaluated models, the XGBoost Regressor (XGBR) demonstrated the strongest predictive performance after the removal of higher-scale project outliers, achieving an RMSE of ₱70,031.88, a MAPE of 0.87%, and an R^2 of 0.9970. These results indicate that the model was able to closely estimate the actual project costs within the scope of the available dataset. With feature importance, the model demonstrated that flow rate was the most influential feature in predicting project cost, which is consistent with the relationship between the capacity and the overall project requirements.

Overall, the developed XGBR model demonstrates potential as a decision-support tool for preliminary budgetary estimation of wastewater treatment plant projects. Rather than replacing detailed engineering design and commercial costing, the model may be used to provide an initial cost estimate before the completion of a full technical proposal. This can help the design and commercial teams prioritize inquiries, assess project feasibility, and reduce the time required to prepare preliminary budgetary estimates.

The model's applicability is currently limited by the size and distribution of the available dataset, particularly the limited representation of higher-scale projects. Further data acquisition for larger and more diverse dataset is therefore recommended before applying the model to projects outside the range of the training data.